Hydraulic Circuit

Symbol representation

Simple Circuits

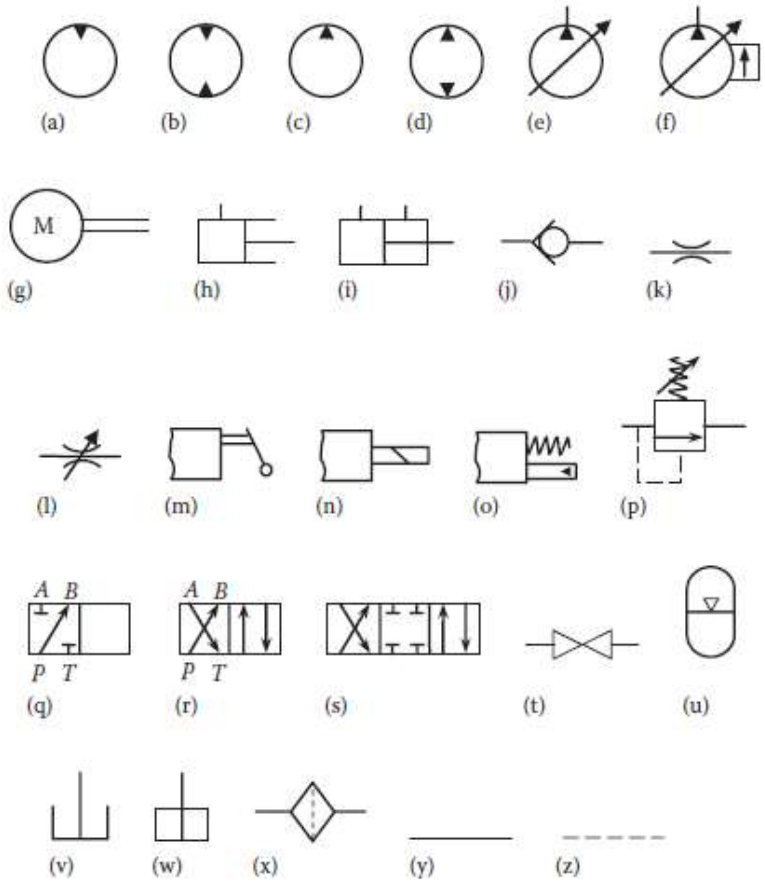
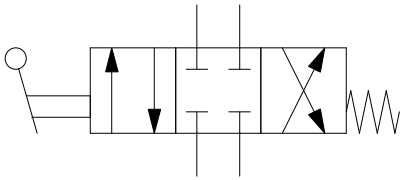


Figure 9.69 Typical graphic symbols used in hydraulic circuit diagrams: (a) motor, (b) reversible motor, (c) pump, (d) reversible pump, (e) variable displacement pump, (f) pressure-compensated variable displacement pump, (g) electric motor, (h) single-acting cylinder, (i) double-acting cylinder, (j) ball-and-seat check valve, (k) fixed orifice, (l) variable flow orifice, (m) manual valve, (n) solenoid-actuated valve, (o) spring-centered pilot-controlled valve, (p) relief valve (adjustable and pressure-operated), (q) two-way spool valve, (r) four-way spool valve, (s) three-position four-way valve, (t) manual shut-off valve, (u) accumulator, (v) vented reservoir, (w) pressurized reservoir, (x) filter, (y) main fluid line, and (z) pilot line.

Directional Control Valves (DCV)

- Control flow **path and direction**
- **Common types:**
  - 2/2 (2 ports, 2 position)
  - 3/2 (3 ports, 2 positions)
  - 4/2 (4 ports, 2 positions)
  - 4/3 (4 ports, 3 positions)

  First number: Number of ports or ways
  Second Number: Number of positions

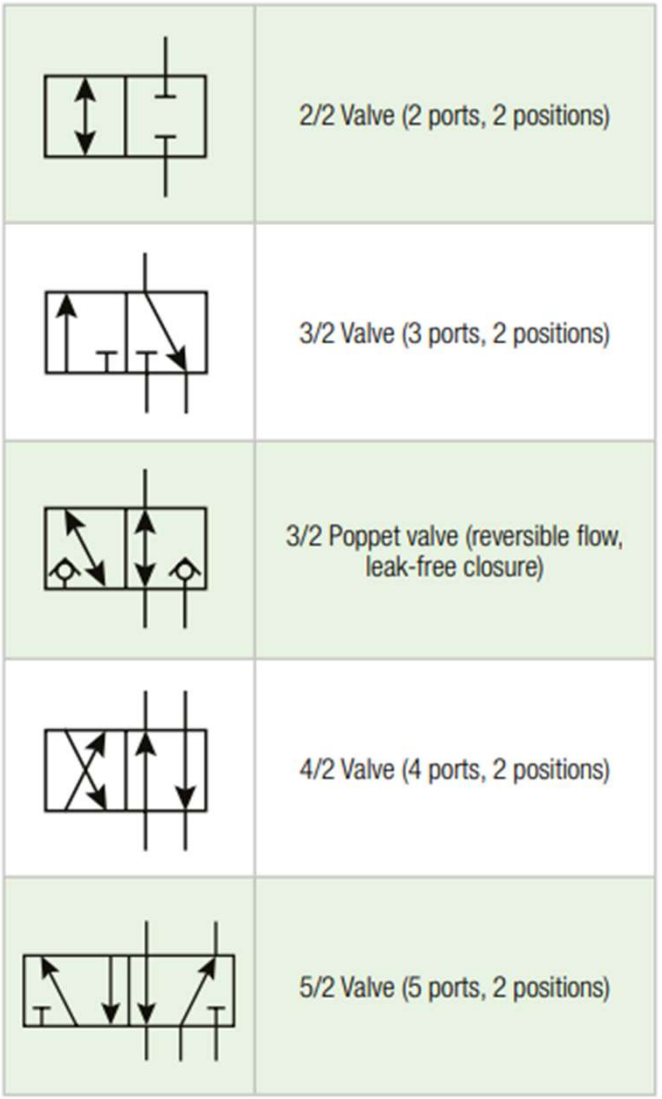


4/3 Valve

Ports: **P (pressure), T (tank), A/B (actuator)**

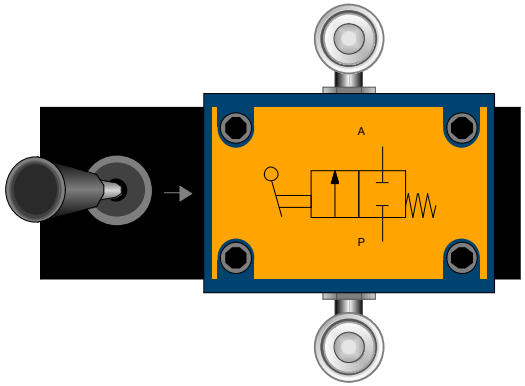
- Valve moves to other positions by actuating it

Actuation Methods: manual, solenoid, spring…

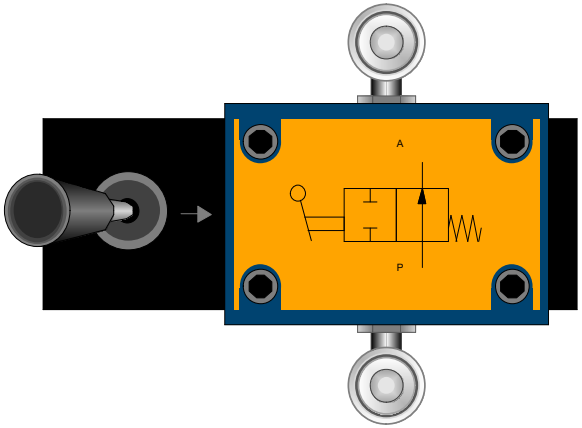


Example: 2/2 (2 ports, 2 positions)

A 2/2 valve is **a two-port, two-position** valve that functions as a simple on/off switch to control the flow of fluid in a system.
The first "2" refers to the two ports (an inlet and an outlet), and the second "2" refers to the two positions (open or closed)

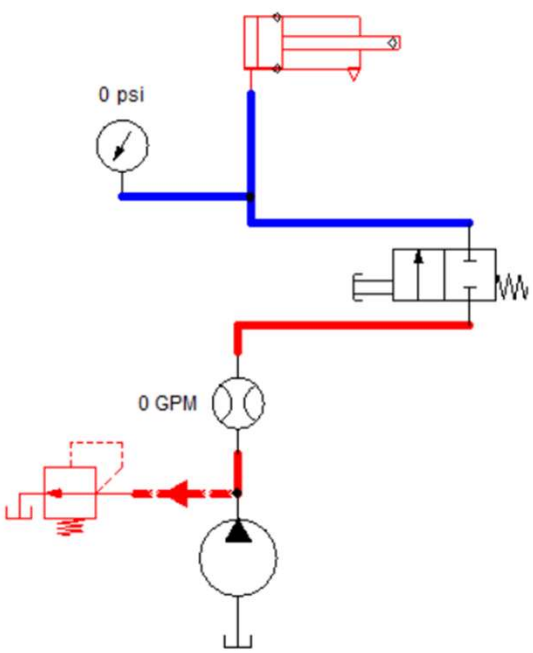


Normally Closed (N.C): The valve is closed by default and doesn't allow fluid to flow.
When valve is actuated, it moves the valve mechanism to an open position, allowing fluid to flow.

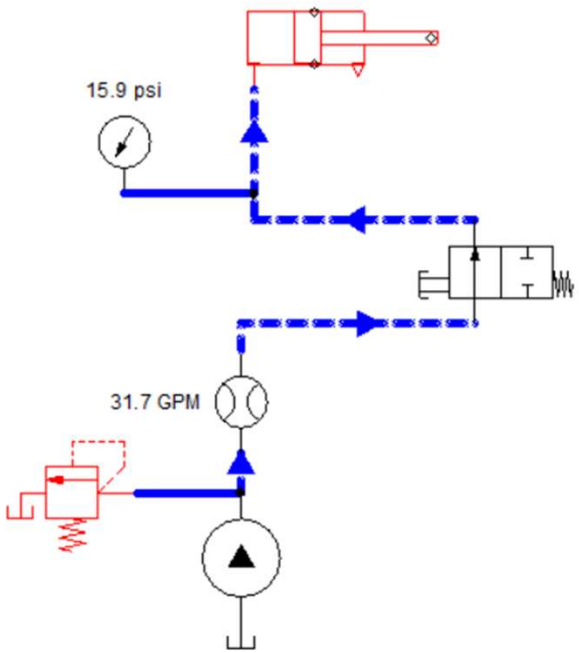


Normally Open (N.O): The valve is open by default and allow fluid to flow.
When valve is actuated, it moves the valve mechanism to a closed position, stops fluid flow.

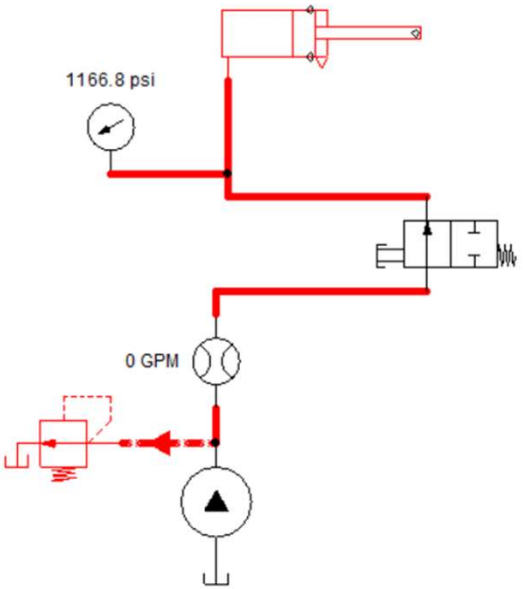
2/2 Directional Valve Normally Closed (NC) Type



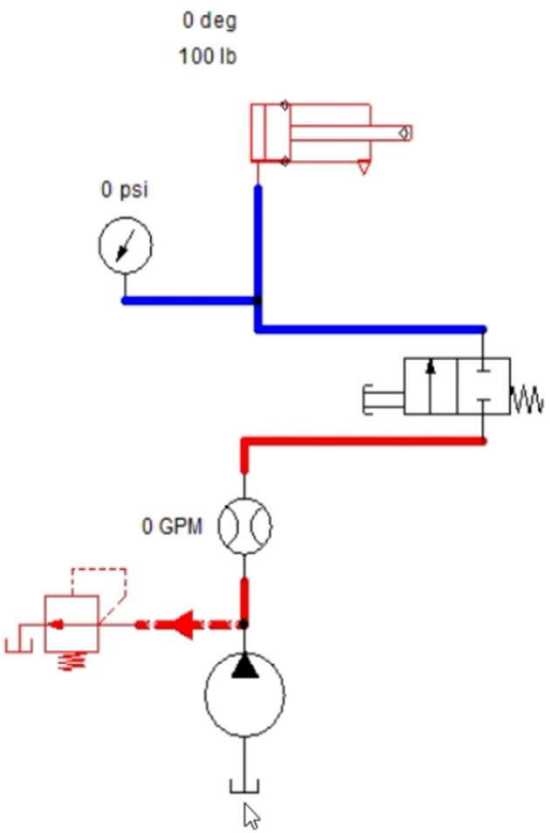
Not actuated: Valve is closed by default (The path is blocked and no flow)



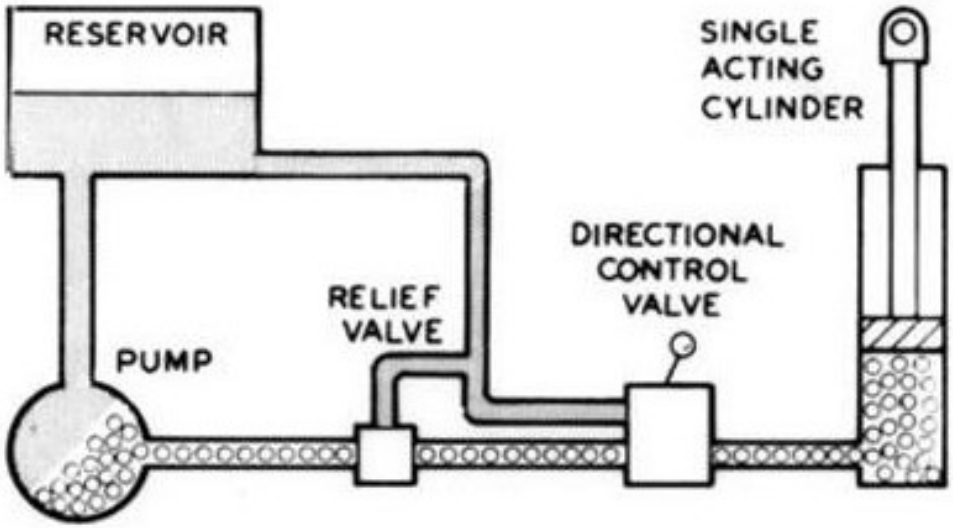
Actuated: Valve is open and allows flow to the cylinder

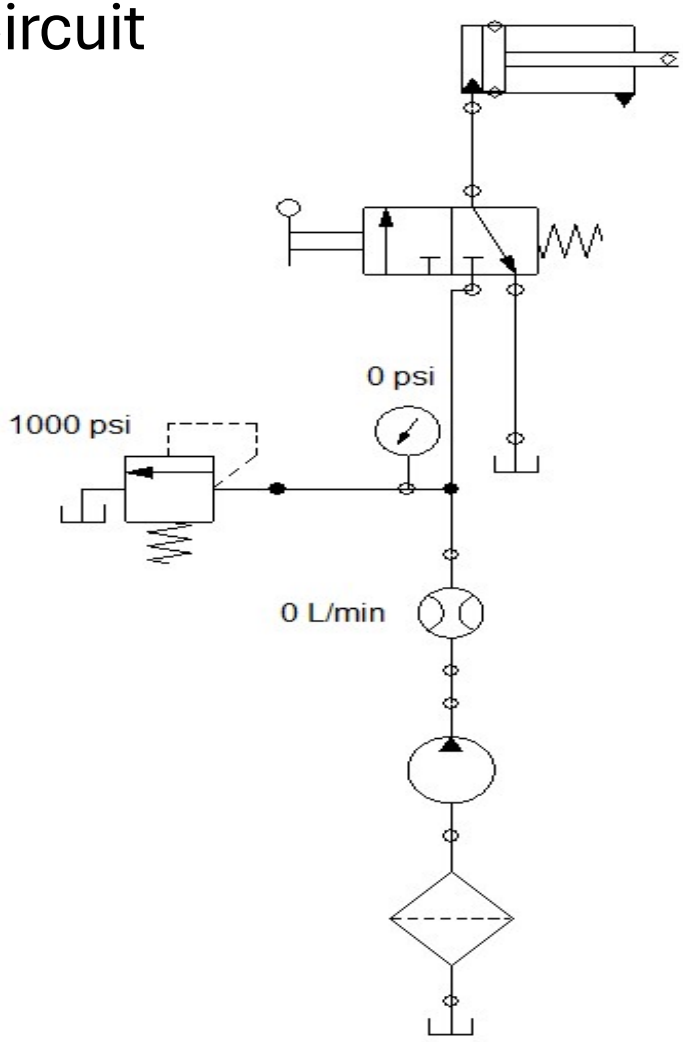


Return to tank through relief valve: Valve is open and allows flow back to the tank

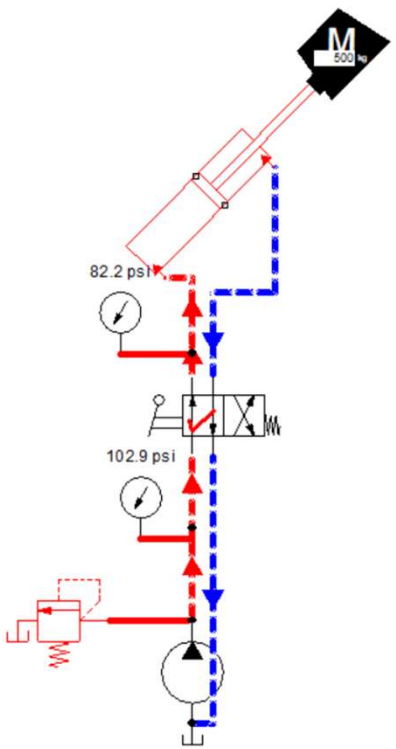

0 deg
100 lb
0 psi
0 GPM

Single Acting Cylinder Circuit

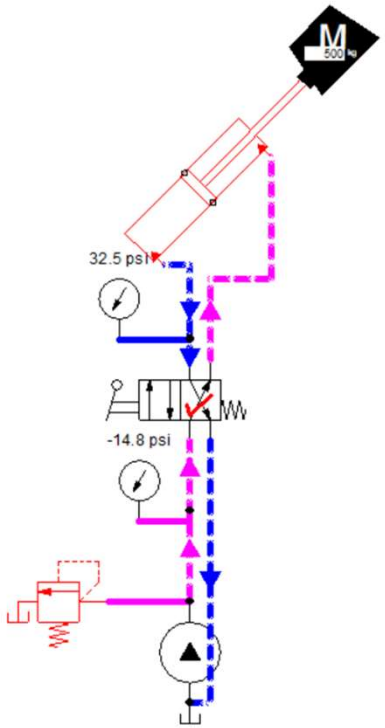




Double-Acting Cylinder Circuit



Extension (Power Stroke)



Retraction (Return Stroke)